

Indian Statistical Institute
M.Tech. (CS), First Year, Mid-Sem of Second Semester Examination, 2023-24
Database Management Systems

Full Marks: 30

Date: 02-03-2024

Time: 2 Hours

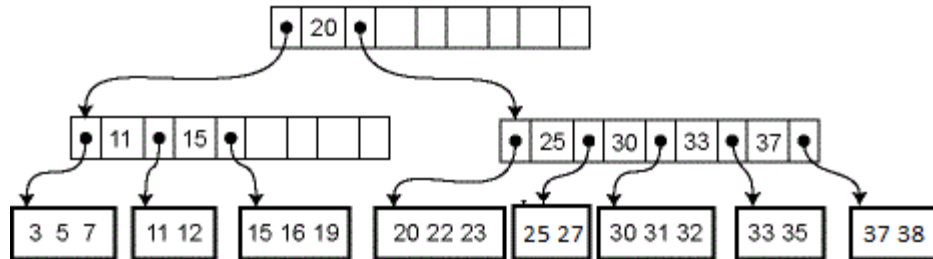
Answer any *three* of the following questions

$3 \times 10 = 30$

1. (a) Cite an example where the natural join and Cartesian product operations generate the same relation as output from the pair of relations on which they are applied.
(b) Prove or disprove the statement: "Given a pair of relations R_1 and R_2 , the division operation $R_1 \div R_2$ monotonically increases (in size) with respect to R_1 but monotonically decreases (in size) with respect to R_2 ".
(c) Let there be a pair of relations $R_1(X, Y)$ and $R_2(Y, Z)$ having t_1 and t_2 ($t_1 > t_2 > 0$) tuples, respectively. If the number of tuples appearing in relation resulting from the operation $R_1 \bowtie R_2$ is n , then what will be the number of tuples in relation resulting from the operation $((R_1 \bowtie R_2) - (R_1 \bowtie R_2)) \cup ((R_1 \bowtie R_2) - (R_1 \bowtie R_2))$?

2+4+4

2. (a) Consider the B-tree shown below. What is the order of the given B-tree? Delete the elements one by one from the B-tree that fall within the range $[17, 22]$ in decreasing order and show the reconstruction of the residual tree in each case.



- (b) Let there be an attribute pertaining to names of people within a database. Entries to this attribute get expanded continuously and as and when such an expansion happens the expanded list is getting returned to the user (in no particular order). What is the best indexing strategy that can be adopted by the database for the said attribute? Justify your answer.

(1+6)+3

3. Let there be a relation $R = \langle A, B, C, D, E \rangle$ having all the attributes as atomic and holding the functional dependencies $FD = \{A \rightarrow B, A \rightarrow C, B \rightarrow D, D \rightarrow E\}$. Answer all the following questions with proper justifications.
(a) What is the candidate key of R ?
(b) What is the highest normal form that R satisfies?

- (c) Decompose R such that every decomposed subrelation of R satisfies the Boyce-Codd normal form.
- (d) Is your decomposition lossless?
- (e) Is your decomposition dependency preserving?

1+1+4+2+2

4. (a) Consider a pair of relations $R_1 = \langle \underline{X}, Y, Z \rangle$ and $R_2 = \langle X, Y, Z \rangle$ having 720 and 5040 tuples, respectively. The attribute X is a primary key for R_1 but not for R_2 . If the number of distinct entries of $X \in [0, 7199]$ in both the tables are same, estimate the size of the following queries.

(i) $(\sigma_{X > 99}(R_1)) \cap (\sigma_{X=0}(R_2))$

(ii) $(\sigma_{X > 99}(R_1)) \cup (\sigma_{X=0}(R_2))$

- (b) Let there be a relation $R = \langle \underline{A}, B \rangle$ having 100 tuples and A as the primary key. If the number of distinct entries in B is 10, estimate the size of the following query.

$$R \bowtie \pi_B(R)$$

(3+3)+4